Automated Brain Cell Detection Using Deep Learning Techniques

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CERTIFICATE OF

This is to certify that the project titled **Automated Brain Cell Detection Using Deep Learning Techniques** is a record of the bonafide work done by **K Veera Vamsiswar Reddy** (*Reg. No. 215890028*) submitted in partial fulfilment of the requirements for the award of the Degree of Bachelor of Technology (B.Tech) in CSE AI of Manipal Institute of Technology Bengaluru, Karnataka, (A Constituent Institute of Manipal Academy of Higher Education), during the academic year 2025-2026.

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ABSTRACT

Artificial intelligence has made great advances in the field of medical image analysis. and biomedical engineering as it can help researchers understand complex cellular structures. High-resolution microscopy images of the brain reveal densely packed and overlapping cells. making manual detection and segmentation a challenging and time-consuming task. Latest developments In deep learning and computer vision, automated systems have been able to accurately detect and segmentation of microcellular structures. The main target of this project is to develop Automated framework for detection and segmentation of brain cells by advanced deep learning

The proposed system integrates multiple state-of-the-art segmentation and detection models such as SAM, CA-Net, Mask R-CNN, SAHI, DETR, BDA, MEDIAR, and YOLOv8 in one unified platform. The workflow starts with image acquisition and preprocessing, followed by model selection and performance using a backend framework. High-resolution microscopy images are analyzed by using deep learning algorithms for detection of cellular structure from segmentation masks and bounding boxes. It also offers a graphical user interface for image upload, visualization, and comparison.The system also includes a graphical user interface for image upload, visualization, and comparative model comparisons.

The first implementation demonstrates that deep learning models can successfully identify and segment cellular structures from microscopy images with higher accuracy and efficiency than manual analysis. YOLOv8 and other models offer fast object detection, while segmentation models like SAM and Mask R-CNN offer detailed information about cell boundaries. The system also supports a comparative evaluation of different approaches, helping researchers to identify adequate models for particular biomedical imaging tasks. These findings point to the potential of deep learning to improve automation of medical image analysis applications. In summary, the project provides a scalable and flexible framework for automated brain cell detection by employing deep learning techniques.

The proposed system reduces the manual effort and enhances analysis efficiency and facilitates biomedical research using high-resolution microscope pictures. The framework can be extended in the future with other datasets, advanced models, and real-time deployment capabilities for medical applications. cellular structures from microscopy images with improved accuracy and efficiency compared to manual analysis. Models such as YOLOv8 provide fast object detection capabilities, while segmentation models such as SAM and Mask R-CNN generate detailed cell boundary information. The system also enables comparative evaluation of different approaches, helping researchers identify suitable models for specific biomedical imaging tasks.

These results indicate the potential of deep learning in improving automation within medical image analysis applications. In conclusion, the project provides a scalable and flexible framework for automated brain cell detection by employing deep learning techniques. The proposed system reduces the manual effort and improves enhances analysis efficiency and facilitates biomedical research using high-resolution microscopy pictures. The framework can be extended in the future with other datasets, advanced models, and real-time deployment capabilities for medical applications.

Software/Tools Used: Python OpenCV PyTorch TensorFlow React Flask Google Colab**,** and YOLOv8 .

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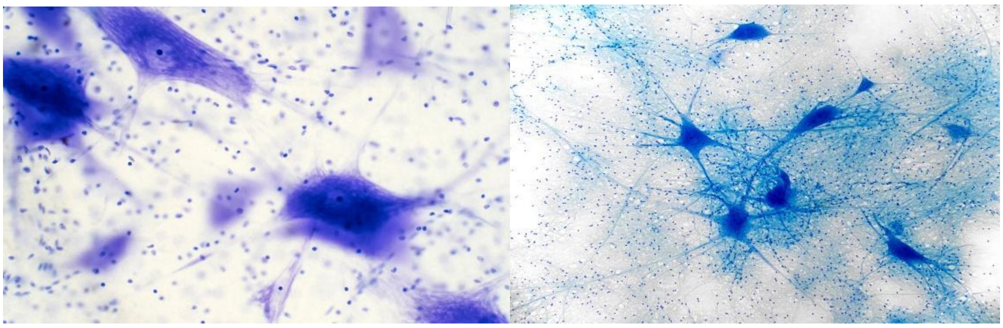
CHAPTER 1
INTRODUCTION

1.1 Introduction

Medical image analysis has become one of the most important research areas in artificial intelligence. and computer vision due to its applications in biomedical and healthcare systems. Microscopy-based brain image analysis helps researchers understand cellular structures, brain tissue organization, and complex biological patterns. Manual analysis of microscopic brain images is difficult because of overlapping cells, varying intensity levels, irregular boundaries, and dense cellular arrangements. These challenges increase the need for automated systems capable of performing accurate cell detection and segmentation.

Recent advancements in deep learning have significantly improved the performance of image segmentation and object detection systems. State-of-the-art models such as SAM, Mask R-CNN, YOLOv8, MEDIAR, and BDA provide powerful techniques for analyzing complex microscopy. images. This project focuses on developing an automated framework for brain cell detection. using advanced deep learning techniques. The proposed system integrates multiple segmentation and detection models within a unified platform to improve analysis efficiency and support comparative evaluation of different approaches.

The project also includes a Graphical User Interface (GUI) for image upload and result visualization. along with backend integration for model execution. The system is designed to process high-resolution brain microscopy images and generate outputs such as segmentation masks and bounding boxes. The framework aims to reduce manual effort and provide an efficient research-oriented solution for automated brain image analysis.



(a) (b)

Figure 1.1 Sample Brain Microscopy Images

1.2 Motivation

Brain cell detection and segmentation are essential tasks in biomedical image analysis and neurology. research. Traditional manual analysis methods are time-consuming and often affected by human error. especially when dealing with large volumes of high-resolution microscopy images. Dense cellular Structures and overlapping cells make accurate identification difficult using conventional images. processing techniques.

Several existing systems focus only on a single segmentation or detection model, limiting them. adaptability across different datasets and imaging conditions. Some methods fail to accurately Identify cell boundaries, while others require extensive preprocessing and computational resources. These limitations motivated the development of a unified framework capable of integrating multiple deep learning models for automated brain cell detection and segmentation.

The proposed project aims to simplify the analysis process by combining state-of-the-art segmentation and detection architectures, all in a single platform. Models like BDA, MEDIAR, SAM, CA-Net, Accuracy is improved using Mask R-CNN and YOLOv8, and comparative evaluation is supported. The The system also provides visualization via a user-friendly GUI, enabling researchers to carry out efficient image analysis without extensive manual intervention.

1.3 Aims of the Work

The primary objective of this project is to develop an automated deep learning-based framework for detection and segmentation of brain cells in high-resolution microscopy images. The system has a goal to combine several state-of-the-art segmentation and detection models into one platform for efficient analysis of biomedical images. The project also aims to put in place a single workflow for image preprocessing, model execution, post-processing, and visualization of the results. Models like SAM, Mask R-CNN, CA-Net, segmentation, and object detection are performed using BDA, MEDIAR, and YOLOv8 tasks. The framework is intended to facilitate the comparative evaluation of different models under the same experimental condition. A secondary objective is the development of a GUI for image upload and visualization, backend integration with Flask, and using preprocessing techniques such as normalization, tiling, and cutback. The project also seeks to reduce the manual effort and enhance the efficiency of brain image analysis by automation.

The project also focuses on implementing a unified workflow for image preprocessing, model execution, post-processing, and result visualization. Models such as SAM, Mask R-CNN, CA-Net, BDA, MEDIAR, and YOLOv8 are incorporated for performing segmentation and object detection. tasks. The framework is designed to support comparative evaluation of different models under the same experimental conditions.

Secondary objectives include the development of a GUI for image upload and visualization, backend integration using Flask, and implementation of preprocessing techniques such as normalization, tiling, and slicing. The project also aims to reduce manual effort and improve the efficiency of brain image analysis through automation.

1.4 Target Specifications

The proposed system is intended to handle high-resolution brain microscopy images with deep learning-based approaches for segmentation and detection. The framework is designed with multiple state-the state-of-the-art models that generate segmentation masks, contour maps, and bounding boxes for cellular architectures.

The system consists of a frontend GUI for interaction with the users and a backend processing framework for Deep learning models deployment. Implementation with Google Colab and GPU acceleration and experimental analysis. The framework is designed to provide efficient visualization, model comparison and scalable experimentation for biomedical image analysis tasks.

The output of the project is an automated and flexible platform to help researchers in brain cell detection and segmentation tasks. The framework aims to enhance accuracy, decrease processing time and simplify experimental workflows in medical image analysis research.

Table 1.1 Deep Learning Models Used in the Proposed System.

Category	**Models Used**
Segmentation Models	SAM, CA-Net, Mask R-CNN
Detection Models	SAHI, DETR, FreeSolo
Cell Detection Models	BDA, MEDIAR, YOLOv8

1.5 Project Timeline

The project work is planned for a duration of four months including literature review, dataset understanding, carrying out experiments, writing reports. First stages concentrate on understanding deep learning models and biomedical imaging techniques then development and implementation of the framework suggested.

The implementation activities include model integration, preprocessing, backend execution and output visualization. In the course of the comparative analysis and documentation, the last stages of the project. The schedule assures systematic completion of project objectives within the planned duration.

Table 1.2 Project Work Schedule.

Month	**Work Carried Out**
January 2026	Literature review and dataset understanding
February 2026	GUI development and integration of SoA models
March 2026	Implementation of BDA, MEDIAR, and YOLOv8
April 2026	Result analysis and report preparation

1.6 Organization of the Project Report

The project report is organized into 5 chapters in different stages of the proposed work.

Chapter 1 introduces the project area, objectives, motivation, and scope of the work. It also discusses the importance of automated brain cell detection using deep learning techniques.

Chapter 2: Background Theory and Literature Review Relevant to Deep Learning-Based segmentation and detection models. Current approaches, recent progress and limitations current techniques are discussed in detail .

Chapter 3 discusses the proposed methodology and the workflow architecture. Preprocessing techniques, model integration and implementation details.

Chapter 4 discusses experimental setup, analysis of results, comparison of models and discussion of extracted outputs.

The work is concluded with chapter 5 summarizing the project findings and future scope and potential improvements.

CHAPTER 2
LITERATURE REVIEW

2.1 Introduction

This chapter discusses the background theory and literature related to deep learning-based brains. cell detection and segmentation techniques. Recent advancements in artificial intelligence and Computer vision has significantly improved biomedical image analysis, especially in microscopy. image processing. Various segmentation and object detection models have been proposed for identifying cellular structures from high-resolution brain microscopy images. This chapter reviews the present state of research, theoretical concepts, existing methodologies, and limitations of current approaches. It also discusses state-of-the-art segmentation and detection models used in the proposed framework.

2.2 Introduction to the Project Area

Brain microscopy image analysis is an important research area in biomedical engineering and medical image processing. Detecting and segmenting brain cells from microscopy images helps researchers Analyze tissue structures, study neurological patterns, and understand complex cellular behavior. However, microscopy images often contain overlapping cells, irregular boundaries, and dense cellular regions, making manual analysis extremely difficult.

Traditional image processing methods are limited in handling such complex cellular structures. The development of deep learning techniques, automated segmentation, and object detection systems have become more efficient. Deep learning models can automatically extract meaningful features. from images and identify complex patterns that are difficult to detect manually. This project focuses on integrating multiple advanced deep learning models within a unified framework for automated brain cell detection and segmentation.

2.3 Literature Review

Recent developments in deep learning have introduced several advanced models for image segmentation. and object detection. Convolutional Neural Networks (CNNs) and transformer-based architectures are widely used in medical image analysis due to their ability to process high-dimensional image data efficiently. Models such as SAM, Mask R-CNN, CA-Net, DETR, SAHI, BDA, MEDIAR, and YOLOv8 has demonstrated strong performance in segmentation and detection tasks.

The Segment Anything Model (SAM) is a foundational segmentation model capable of generating segmentation masks for different image regions without task-specific retraining. Mask R-CNN is an instance segmentation architecture that performs object detection and mask generation simultaneously. CA-Net uses attention mechanisms to improve segmentation accuracy by focusing on important image regions.

Object detection models such as DETR and YOLOv8 provide efficient detection of objects using transformer architectures and real-time detection frameworks. SAHI improves small objects. detection performance by slicing large images into smaller patches before inference. MEDIAR focuses on high-resolution microscopy image segmentation through tiled image processing, while BDA is designed for accurate cell boundary analysis and contour extraction.

These models have drastically advanced biomedical image analysis, but most existing systems focus on one architecture only. Comparison between different segmentation and detection methods is often limited. This leads to the need for a unified framework capable of integrating multiple deep learning models for systematic experimentation and evaluation

Table 2.1 Literature Survey Summary

Model	**Technique**	**Major Advantage**	**Limitation**
SAM	Segmentation	Zero-shot segmentation	High computational requirement
Mask R-CNN	Instance Segmentation	Accurate mask generation	Slower inference
CA-Net	Attention Segmentation	Improved boundary focus	Complex architecture
DETR	Transformer Detection	End-to-end detection	Requires larger training data
SAHI	Sliced Inference	Better small object detection	Increased processing time
BDA	Cell Boundary Detection	Accurate contour extraction	Sensitive to image quality
MEDIAR	Tiled Segmentation	Handles high-resolution images	Patch reconstruction complexity
YOLOv8	Real-time Detection	Fast inference speed	Limited boundary segmentation

2.3.1 Evaluation of important research papers

SAM: Segment Anything Model

Segment Anything Model, also known as SAM, was proposed by Kirillov et al. as a general segmentation model that can generate high quality masks for different types of images[6].

In this project, SAM is considered because it supports flexible segmentation without requiring task-specific training for every new image type. Its ability to generate segmentation masks makes it useful for identifying cellular regions in brain microscopy images. However, SAM requires higher computational resources and may not always be optimized for domain-specific biomedical images with no further adaptation.

YOLO: You Only Look Once
Redmon et al. introduced YOLO as a single-stage detector for real-time object detection [8]. YOLO detects YOLO detections. objects directly from images using a single neural network, making it faster compared to many traditional detection pipelines. In this project, YOLOv8 is chosen for cell detection due to it fast inference speed and ability to generate bounding boxes around detected cellular regions. The YOLOv8-F-T repository also includes output images and notebook implementation workflow, making it useful for demonstrating the detection pipeline in the project.

MEDIAR
Lee et al. proposed MEDIAR for multi-modality microscopy image segmentation [7]. MEDIAR is important for this project because it is specifically designed for microscopy-based segmentation tasks. It processes microscopy images by focusing on cell recognition and instance distinction. making it suitable for high-resolution brain microscopy images. In the proposed project, MEDIAR is considered useful because it can handle large images through tiled processing and reconstruct the final segmented output.

BDA
Upschulte et al. proposed contour proposal networks for biomedical instance segmentation [10]. BDA is relevant to this project because it focuses on contour-based detection and boundary extraction, which is very important in brain cell segmentation. The BDA_Cell-detection repository contains input folders. output folders, model execution scripts, and dependency details required for local execution. The The repository also supports running a single image using python main.py input_image_path. output_image_path, making it suitable for testing contour-based cell detection outputs.

Mask R-CNN
Mask R-CNN is an instance segmentation model that extends object detection by generating pixel-level masks for each detected object [16]. It is useful in this project because brain cell detection requires not only locating cells but also identifying their boundaries. Mask R-CNN provides detailed segmentation outputs, making it suitable for comparison with faster detection models such as YOLOv8. However, it generally requires more computation and may have slower inference. compared to object detection models.

DETR
DETR, or Detection Transformer, introduced transformer-based object detection using an end-to-end architecture [12]. It removes the need for many hand-designed detection components and uses attention mechanisms for object localization. In this project, DETR is included as a detection model because it represents recent transformer-based developments in computer vision. It helps Compare conventional detection models with transformer-based detection approaches.

2.4 Present State and Recent Developments

Deep learning-based medical image analysis has experienced rapid growth due to improvements. in computational resources and neural network architectures. Modern segmentation models are capable of generating highly detailed masks even in complex microscopy images. Transformer based architectures have further improved object detection performance through attention mechanisms and end-to-end learning approaches.

Recent research focuses on improving segmentation accuracy in densely packed cellular structures. Models such as MEDIAR and BDA specifically address microscopy image analysis by introducing tiled segmentation and contour-based detection techniques. Similarly, YOLOv8 provides efficient real-time object detection for identifying cellular regions with high speed and accuracy.

Another important development is the integration of graphical interfaces and backend frameworks for improving usability and experimentation. Researchers are increasingly focusing on building unified systems capable of handling multiple segmentation and detection models under the same environment. This project follows a similar direction by integrating multiple deep learning approaches into a single framework.

2.5 Background Theory

Deep learning models are widely used in image analysis because of their ability to automatically extract features from raw image data. Convolutional Neural Networks (CNNs) use convolution layers to identify spatial patterns and image features. Segmentation models generate pixel-level outputs to identify regions of interest, while object detection models generate bounding boxes around detected objects.

Transformer-based models such as DETR use self-attention mechanisms to understand relationships between different image regions. Attention-based segmentation models improve feature extraction by focusing on important areas within the image. Models such as SAM and CA-Net use advanced feature representation methods for improved segmentation accuracy.

Image preprocessing techniques also play an important role in microscopy image analysis. Operations such as resizing, normalization, slicing, and tiling improve model performance and computational efficiency. Post-processing techniques such as contour extraction and mask reconstruction help refine detection and segmentation outputs.

Figure 2.1 Classification of Deep Learning Models Used

Segmentation Models	**Detection Models**	**Cell Detection Models**
SAM	SAHI	BDA
CA-Net	DETR	MEDIAR
Mask R-CNN	FreeSolo	YOLOv8

2.6 Summarized Outcome of the Literature Review

The literature review indicates that deep learning models have significantly improved the performance of medical image segmentation and object detection tasks. Models such as SAM, Mask R-CNN, BDA, MEDIAR, and YOLOv8 demonstrate strong capabilities in analyzing high-resolution brain microscopy images.

However, existing systems generally focus on individual models rather than unified frameworks. Some models provide accurate segmentation but require high computational resources, while others provide faster detection with limited boundary precision. Comparative evaluation between models is often not supported within a single system.

These observations highlight the importance of developing a unified framework capable of integrating multiple segmentation and detection models. Such a system can help researchers evaluate models. performance efficiently and select suitable approaches for different biomedical imaging tasks.

2.7 Theoretical Discussion and General Analysis

Deep learning-based segmentation and detection systems provide better performance compared to traditional image processing techniques because they learn features directly from image data. Segmentation models are useful for identifying detailed cellular boundaries, while object detection models are more efficient for locating cellular regions quickly.

Theoretical analysis shows that combining multiple deep learning models within a unified framework improves flexibility and experimentation capabilities. Different models perform differently depending on image quality, cellular density, and computational requirements. Therefore, comparative analysis is necessary for selecting suitable models for specific biomedical applications.

The proposed framework addresses this requirement by integrating multiple state-of-the-art models. into a single system. This approach improves usability, supports experimentation, and enhances research productivity in automated brain image analysis.

2.8 Conclusion

This chapter presented the literature review and theoretical background related to deep learning-based brain cell detection and segmentation techniques. Existing segmentation and object detection models such as SAM, Mask R-CNN, DETR, BDA, MEDIAR, and YOLOv8 were discussed along with their advantages and limitations.

The review identified the need for a unified framework capable of integrating multiple deep learning models for comparative evaluation and efficient experimentation. The theoretical discussions and literature analysis provide the foundation for the methodology proposed in the next chapter.

CHAPTER 3
METHODOLOGY

3.1 Introduction

This chapter explains the methodology adopted for the proposed automated brain cell detection system using deep learning techniques. It discusses the overall workflow, preprocessing methods, segmentation and detection models, backend execution, GUI integration, and implementation details. The chapter also describes the tools, technologies, module specifications, and system architecture used for processing high-resolution brain microscopy images. In addition, preliminary outputs and implementation-related analysis are discussed to explain the effectiveness of the proposed framework.

3.2 Proposed Methodology

The proposed system follows a unified deep learning-based methodology for automated brain cell detection and segmentation. The workflow begins with image acquisition, where high-resolution brain microscopy images are provided as input through the graphical interface. The uploaded image is processed and forwarded to the backend framework for model execution.

The system supports multiple segmentation and detection architectures including SAM, CA-Net, Mask R-CNN, SAHI, DETR, BDA, MEDIAR, and YOLOv8. Depending on the selected model, the image undergoes preprocessing operations such as resizing, normalization, tiling, and slicing. These operations improve image quality and computational efficiency during inference.

Segmentation models generate pixel-level masks representing cellular regions, while object detection models generate bounding boxes around detected cells. Post-processing techniques such as contour extraction, mask reconstruction, and noise removal are applied to improve output quality. Finally, the results are visualized within the GUI to support comparative evaluation and analysis.

3.3 Detailed Methodology

The detailed methodology consists of multiple stages including image input, preprocessing, model selection, backend execution, post-processing, and result visualization. Initially, the user uploads a microscopy image through the GUI developed using React. The image is transferred to the Flask backend where preprocessing operations are performed.

For large microscopy images, slicing and tiling techniques are applied to reduce computational complexity. MEDIAR processes images in smaller patches and reconstructs the segmented outputs into complete images. SAHI improves detection performance by dividing high-resolution images into smaller sections before inference.

The processed image is then passed to the selected deep learning model. Segmentation models such as SAM and Mask R-CNN generate segmentation masks, while YOLOv8 and DETR generate objects detection outputs. BDA performs contour-based boundary extraction for identifying detailed cellular structures. The outputs are refined using post-processing techniques and displayed within the graphical user interface.

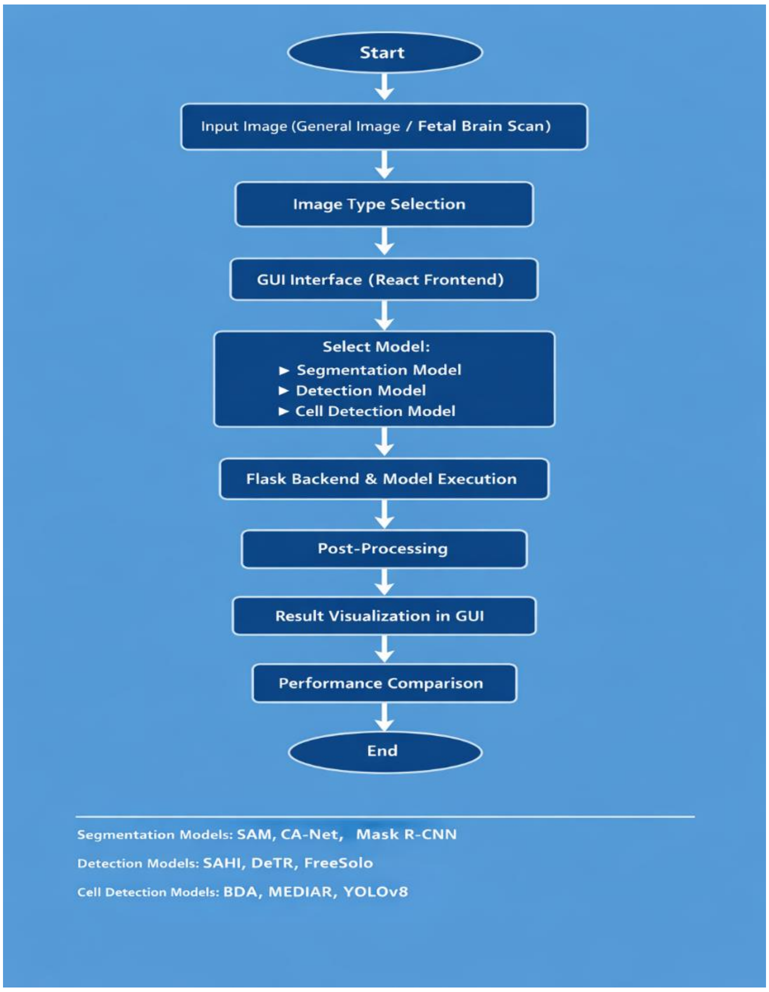


Figure 3.1 System Workflow Architecture

3.4 Assumptions Made

The proposed system assumes that the input microscopy images are of sufficient resolution and quality for deep learning-based analysis. It is assumed that the uploaded images contain visible cellular structures with distinguishable boundaries. The preprocessing techniques are expected to normalize image variations and improve model inference performance.

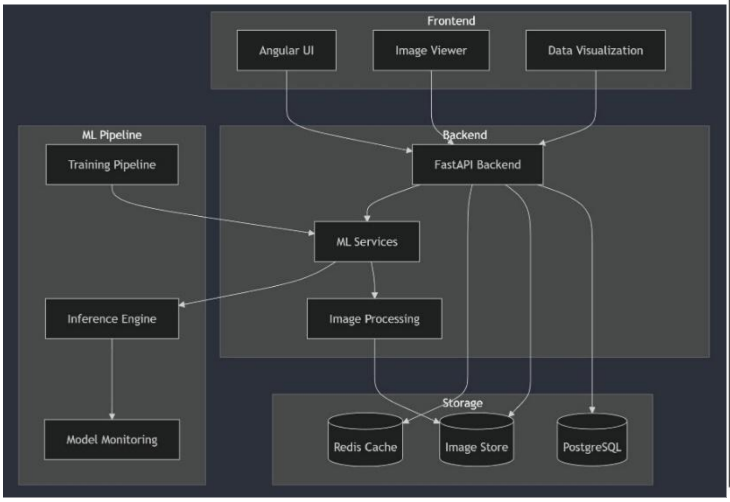
The system also assumes that GPU-enabled environments such as Google Colab are available. for executing computationally intensive deep learning models. Since multiple models are integrated Into the framework, sufficient memory and processing resources are assumed for smooth execution. and result generation.

3.5 Modeling and Design

The proposed framework is based on a modular architecture, which is composed of frontend, backend, preprocessing, model execution, post-processing, and visualization modules. The GUI is designed using React with Flask. support for image upload and model selection The back-end framework used is Flask.to enable communication between the frontend and deep learning models.

The preprocessing module performs resizing, normalization, slicing, and tiling. The model execution module uses segmentation and object detection architectures in order to process the images. The visualization module visualizes the segmentation masks and detection outputs in the interface for analysis.

Scalability and combination with other deep learning models via modular design later stages. It also allows for comparative experimentation by running multiple models. under the same framework.



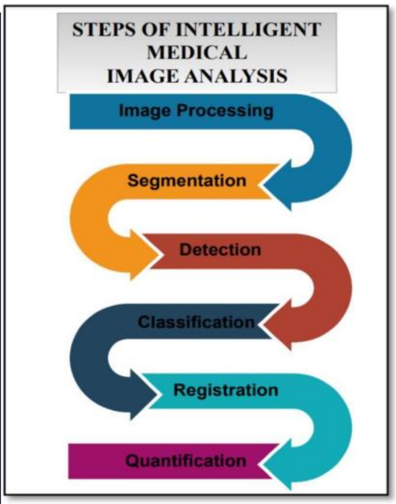


Figure 3.2 GUI and Backend Architecture

3.6 Module Specifications

The system is structured in several modules for easier processing and improved maintainability. The image input module handles image upload operations through the GUI. The preprocessing The module normalizes and resizes the images for the inference. The segmentation module executes models such as SAM, CA-Net, and Mask R-CNN to generate pixel level masks. The detection module uses YOLOv8, DETR, and SAHI to detect the object locations using bounding boxes.

The cell detection module integrates BDA and MEDIAR for Visualization module shows the processed outputs and comparative results on GUI. contour extraction and microscopy image segmentation. The visualization module displays processed outputs and comparative results within the GUI. Each module performs specific tasks independently while maintaining communication through the backend framework.

Table 3.1 Model Specifications of the Proposed System

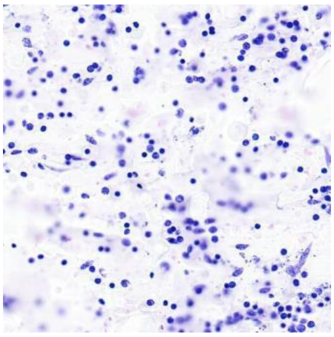
Module	**Function**
Image Input Module	Uploads microscopy images
Preprocessing Module	Performs normalization and resizing
Segmentation Module	Generates segmentation masks
Detection Module	Detects cellular structures
Backend Module	Executes deep learning models
Visualization Module	Displays outputs in GUI

3.6.1 Dataset Description and Sample Images

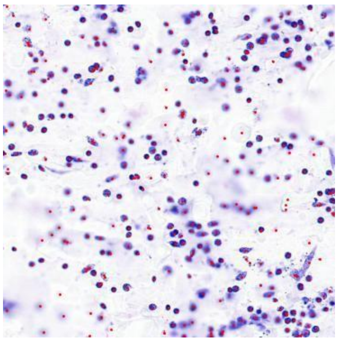
The dataset used in this project consists of high-resolution brain microscopy images containing visible cellular structures. These images are used for evaluating deep learning-based cell detection and segmentation models. The input images contain dense cellular regions, overlapping cell structures, and irregular boundaries, making them suitable for testing segmentation and detection performance.

The dataset includes raw microscopy images along with annotated data used for segmentation and object detection tasks. The images are processed using preprocessing techniques such as normalization, resizing, slicing, and tiling before being passed to deep learning models. These preprocessing operations help improve image quality and computational efficiency during model execution.

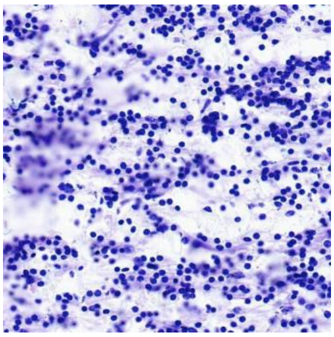
For cell detection, the proposed framework uses YOLOv8-based detection techniques to identify cellular regions efficiently. The generated outputs include bounding box-based detections representing cell locations within microscopy images. For contour-based analysis and segmentation, BDA and MEDIAR models are used to identify cellular boundaries and segmented regions from high-resolution microscopy images. The dataset plays an important role in evaluating the effectiveness of segmentation and detection models under different microscopy conditions. Sample dataset images and generated outputs are included in this report to demonstrate the ability of the proposed framework to process complex brain microscopy images for automated biomedical image analysis.



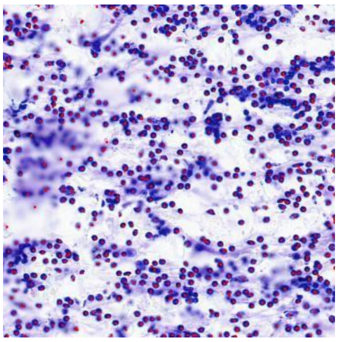
(a) Input image



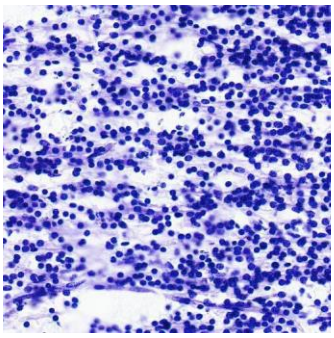
(b) Yolov8/F-T results



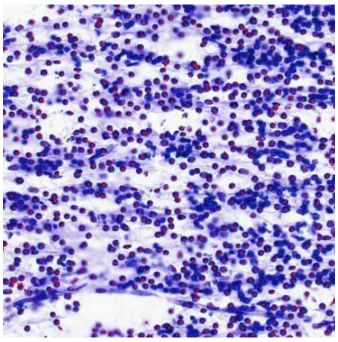
(a) Input image



(b) YOLOv8/F-T results



(a) Input image



(b) YOLOv8/F-T results

Figure 3.3 Sample Brain Microscopy Dataset Images

3.7 Justification for Modules

The preprocessing module is necessary for improving image quality and reducing computational complexity during model execution. Segmentation models are included to provide accurate pixel-level analysis of cellular structures. Detection models are integrated for identifying object locations. efficiently using bounding boxes.

BDA and MEDIAR are specifically selected because they are designed for microscopy images. analysis and contour-based cell detection. YOLOv8 is included for fast object detection and efficient inference. SAM and Mask R-CNN provide high-quality segmentation outputs useful for detailed cellular analysis.

The GUI and backend modules improve usability and simplify interaction with the deep learning framework. The modular architecture also supports scalability and future integration of additional models.

3.8 Tools Used

The project is implemented using Python as the primary programming language. OpenCV is used for image preprocessing and post-processing operations. PyTorch and TensorFlow are used for implementing deep learning architectures and executing segmentation and detection models. YOLOv8 is implemented using the Ultralytics framework for real-time object detection. Flask is used as the backend framework for connecting the frontend with model execution modules. React is used for developing the graphical user interface. Google Collab is used for implementation and GPU-based experimentation. Additional tools such as Matplotlib are used for visualization of results and image outputs. Roboflow is used for image annotation and dataset preparation during model fine-tuning stages.

3.9 Preliminary Result Analysis

Initial implementation results demonstrate successful execution of object detection and segmentation models on microscopy images. YOLOv8 generates bounding boxes around detected cellular regions, while segmentation models produce pixel-level masks representing cellular boundaries. The system successfully processes uploaded images through the backend framework and displays outputs within the graphical interface.

Preliminary observations indicate that segmentation models provide more detailed boundary information, whereas detection models offer faster inference and efficient localization of cellular structures. The comparative outputs generated from different models help analyse the strengths and limitations of each approach. These preliminary results confirm the feasibility of the proposed unified framework for automated brain cell detection and segmentation.

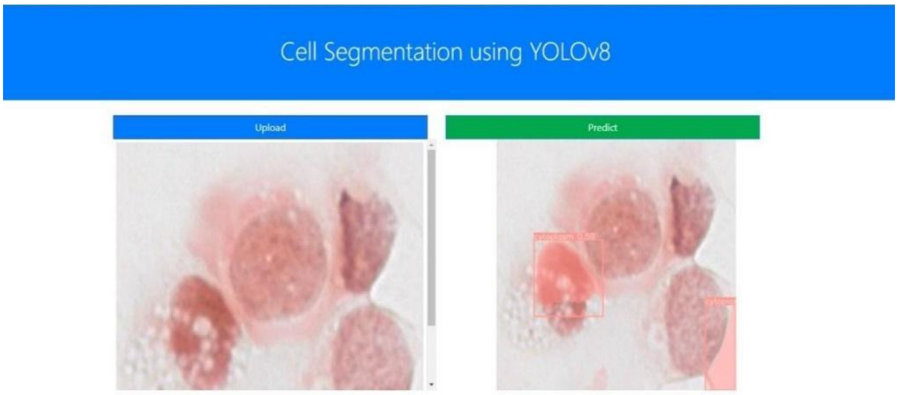


Figure 3.3: Preliminary Detection Output using YOLOv8

3.10 Summary

In this chapter, we discussed the methodology adopted for the proposed automated brain cell detection. system, Workflow architecture, pre-processing methods, segmentation and detection models, Module specifications and implementation tools were explained in detail.

The proposed framework integrates multiple deep learning architectures within a unified environment for efficient biomedical image analysis. Preliminary outputs demonstrate the capability of the system to process microscopy images and generate meaningful detection and segmentation results. The next chapter discusses the experimental setup, result analysis, and performance evaluation of the proposed framework.

CHAPTER 4
RESULTS ANALYSIS

4.1 Introduction

In this chapter we present the experimental results and analysis of the proposed automated brain. cell detection system based on deep learning techniques. The outputs of the segmentation The object detection models are discussed with graphical and tabular representation. Comparison An evaluation of different models is conducted to study their effectiveness in processing high-resolution pictures of brain microscopy the chapter also discusses the importance of the results obtained. limitations encountered during implementation, environmental and social impact of the proposed framework and overall system performance.

4.2 Result Analysis

The proposed framework successfully processes brain microscopy images using multiple deep learning-based segmentation and object detection models. The outputs generated include segmentation. masks, contour maps, and object detection bounding boxes representing detected cellular structures. Models such as YOLOv8 provide fast object localization, while segmentation models such as SAM and Mask R-CNN provides detailed boundary extraction.

The implemented framework allows comparative analysis between segmentation and detection. approaches. Preliminary observations indicate that segmentation models generate more precise cellular boundaries, whereas object detection models provide faster execution and efficient localization. MEDIAR and BDA demonstrate improved performance in handling high resolution. microscopy images through tiled segmentation and contour-based extraction techniques.

The graphical outputs generated during implementation show that the system can effectively Identify cellular regions in complex microscopy images. The framework also enables visualization. of outputs through the GUI, simplifying interpretation and analysis for researchers.

4.2.1 Model-Wise Result Output Analysis

The YOLOv8 model produces detection outputs by marking cellular regions using bounding boxes. boxes or center-based detections. The YOLOv8-F-T repository contains output images, which can be included as detection result samples. These outputs help demonstrate that the model is capable. of identifying cellular regions from brain microscopy images.

BDA produces contour-based outputs by detecting cellular boundaries and representing cells. regions using contour maps. This makes it suitable for visualizing boundary-level details in microscopy images. The BDA repository contains input images and output folders, which can be used to show the difference between raw input images and contour-based outputs.

MEDIAR is used for microscopy image segmentation and performs better when large images are divided into smaller tiles before processing. The segmented tiles are reconstructed to generate the final output. This model is useful for comparing segmentation-based output with YOLOv8 detection output and BDA contour output.

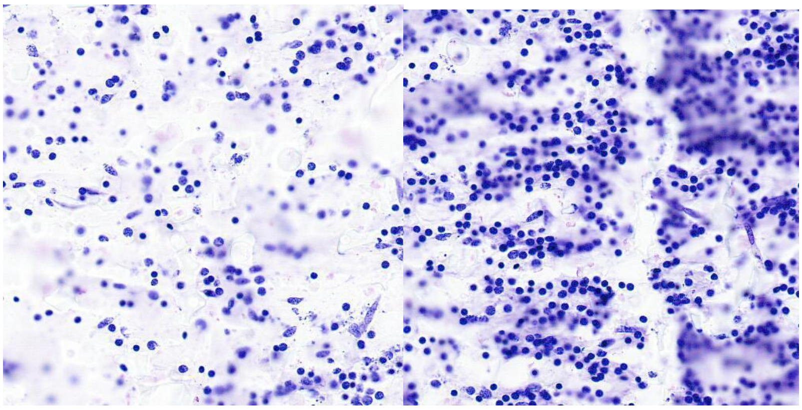


Figure 4.1 YOLOv8 Detection Results

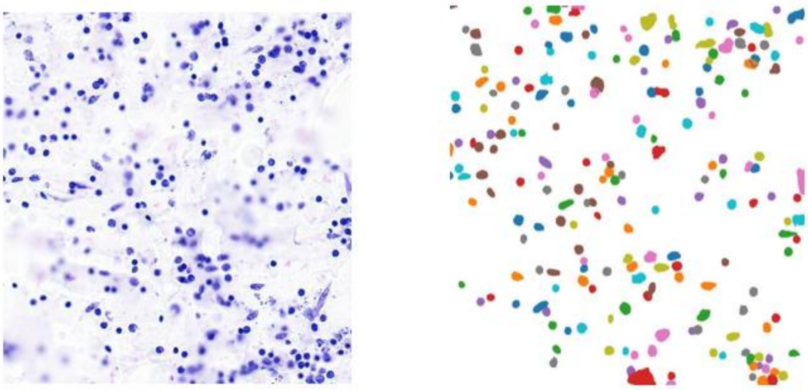


Figure 4.2 BDA Detection Results

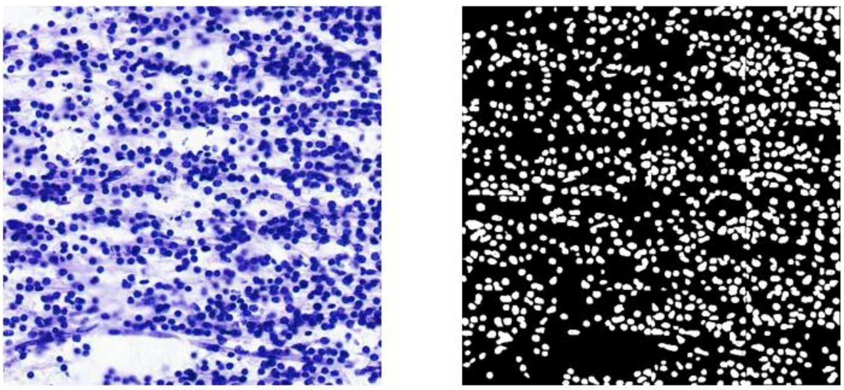


Figure 4.3 MEDIAR Detection Results

Table 4.1 Comparative Analysis of Deep Learning Model

Model	Category	Major Strength	Observation
SAM	Segmentation	Accurate mask generation	Detailed segmentation output
Mask R-CNN	Segmentation	Instance-level segmentation	Better boundary extraction
YOLOv8	Detection	Fast object detection	Faster inference
MEDIAR	Cell Detection	Handles large images	Efficient tiled segmentation
BDA	Cell Detection	Accurate contour analysis	Better cell boundary detection

4.2.2 Comparative Cell Count Analysis

Twenty image tiles were selected from the primary brain microscopy dataset for comparative analysis.

Table 4.2: Set-1 Results of cell count

Image name	GT Count	BDA Count	Yolov8/F-T Count	MEDIAR Count
14730_0_3.jpg	17	13	20	13
14730_1_6.jpg	13	3	12	7
14730_2_4.jpg	18	13	16	17
14730_2_7.jpg	18	17	20	10
14730_4_2.jpg	14	8	14	18
14730_4_5.jpg	14	8	14	13
14730_6_2.jpg	15	13	24	17
14730_1_5.jpg	17	11	15	12
14732_1_7.jpg	12	13	12	14
14732_2_6.jpg	12	13	12	13

Metric	Value
BDA Accuracy (%)	73.73
YoloV8/F-T Accuracy (%)	96.7
MEDIAR Accuracy (%)	80.96

Metric	Value
BDA miss (%)	26.25
YoloV8/F-T miss (%)	3.3
MEDIAR miss (%)	19.04

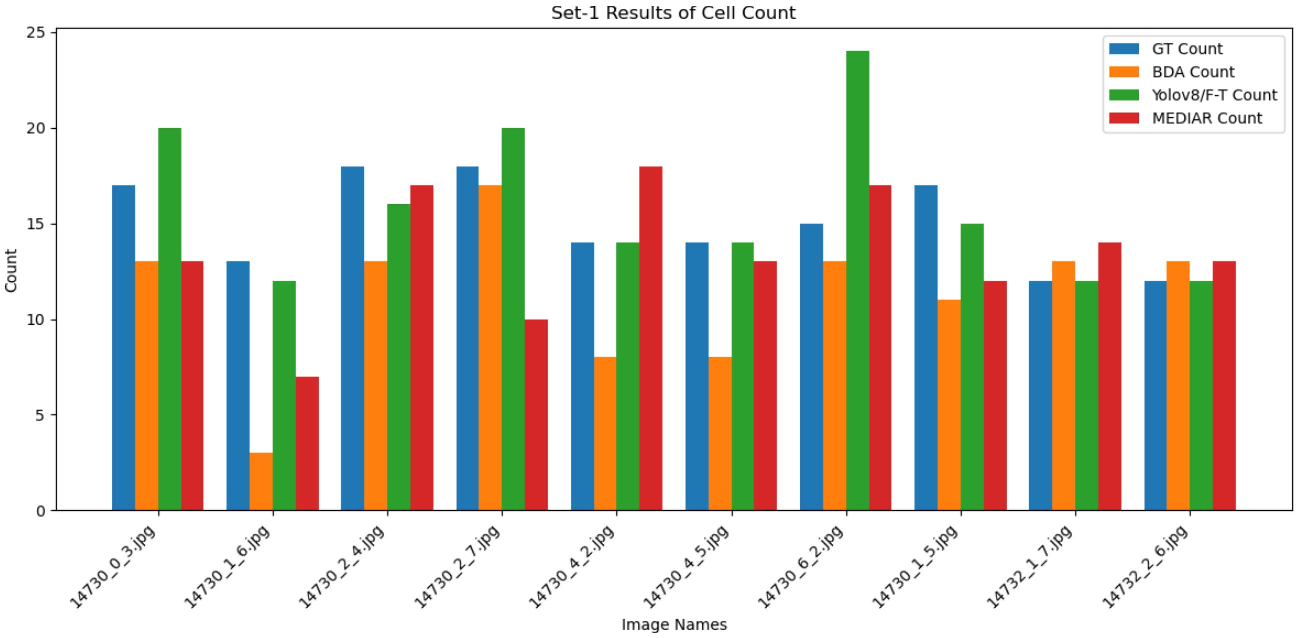


Table 4.3: Set-2 Results of cell count

Image name	GT Count	BDA Count	Yolov8/F-T Count	MEDIAR Count
14732_3_0.jpg	15	11	14	16
14732_4_6.jpg	8	8	10	9
14732_5_1.jpg	8	9	12	14
15188_0_5.jpg	19	14	19	20
15188_1_7.jpg	20	13	24	22
15188_3_7.jpg	13	16	22	18
15188_4_3.jpg	14	7	16	18
15188_4_7.jpg	16	16	15	19
15188_5_6.jpg	13	10	16	17
15188_7_3.jpg	15	11	13	8

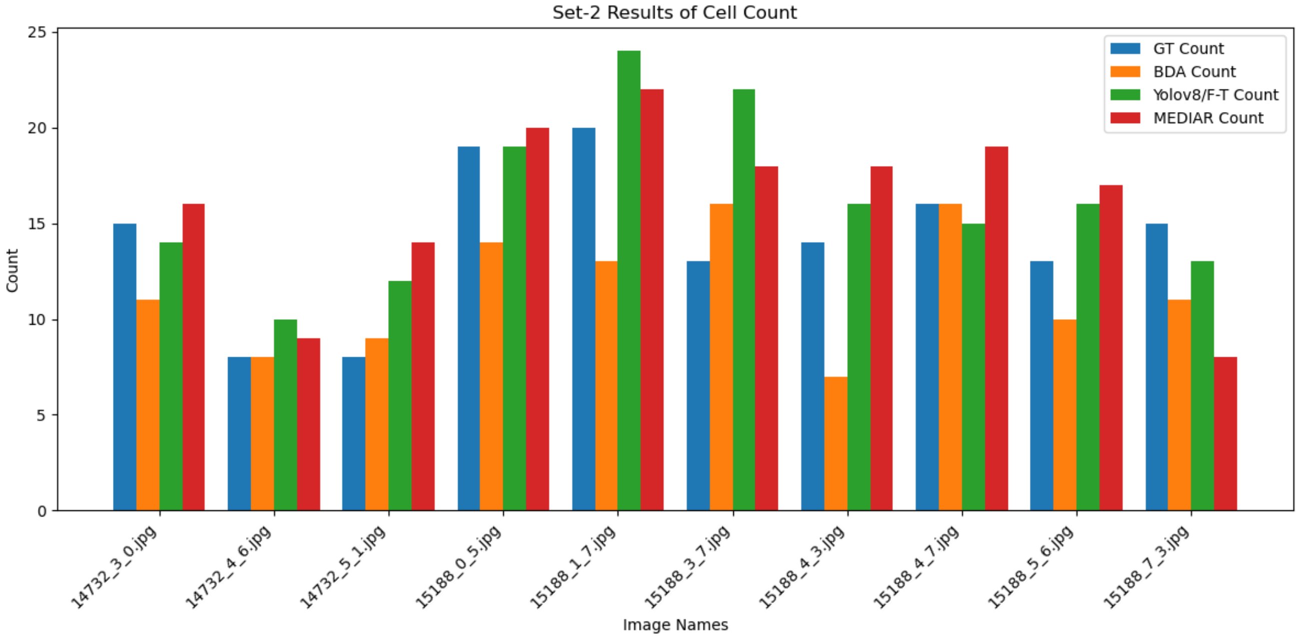
Metric	Value
BDA Accuracy (%)	78.72
YoloV8/F-T Accuracy (%)	93.64
MEDIAR Accuracy (%)	95.8

Metric	Value
BDA miss (%)	21.28
YoloV8/F-T miss (%)	6.36
MEDIAR miss (%)	4.2

After carefully observing both studies, it can be concluded that

1. YOLOv8/F-T avg accuracy is 95.17%.
2. BDA avg accuracy is 76.225%.
3. MEDIAR avg accuracy is 88.38%.

It is observed that MEDIAR and YOLOV8/F-T results are better or equivalent to G-T.



4.3 Explanation of Graphical and Tabular Results

The graphical outputs generated from the implemented models demonstrate the effectiveness of deep learning techniques in brain cell detection and segmentation. Segmentation outputs provide pixel-level identification of cellular structures, enabling detailed analysis of cell boundaries and tissue organization. Detection outputs generated using YOLOv8 highlight cellular regions using bounding boxes for efficient localization.

The table comparison illustrates the strengths and weaknesses of the different models incorporated in the framework. Better precision in segmentation is provided by models such as SAM and Mask R-CNN, while faster execution is provided by YOLOv8, which is good for real-time analysis. MEDIAR and BDA are particularly suitable for high-resolution microscopy images due to their specific pre-processing and contour extraction capabilities. The results confirm that the integration of multiple models into a unified framework provides increased flexibility and allows for comparative experimentation in the field of biomedical image analysis.

4.3.1 Analysis of performance metrics

Performance analysis constitutes an important part of the proposed project, as it helps to assess the effectiveness of each model on brain microscopy images. The main models for performance comparison are YOLOv8, BDA, and MEDIAR. The models are compared for accuracy, type of output, and execution efficiency.

According to the BPT reference report, after fine-tuning, YOLOv8/F-T achieved the highest detection accuracy of 95.17% on the cell dataset. The accuracy of MEDIAR and BDA against the ground truth was 88.38% and 76.225%, respectively. These values indicate that YOLOv8/F-T is preferable for object-level cell detection, while MEDIAR and BDA are helpful for segmentation and contour-based analysis.

YOLOv8/F-T is an efficient model, as it was fine-tuned with annotated cell data. BDA is designed for contour proposal and boundary extraction and is useful for visual analysis of cellular outlines. MEDIAR performs microscopy segmentation by processing the tiled input image and reconstructing the output. Thus, the proposed framework has different contributions from all three models.

Table 4.4 Performance Metrics of Major Cell Detection Models

Model	**Output Type**	**Accuracy**	**Major Observation**
BDA	Contour-based cell segmentation	76.225%	Useful for boundary and contour extraction
MEDIAR	Microscopy cell segmentation	88.38%	Effective for tiled microscopy segmentation
YOLOv8/F-T	Cell detection	95.17%	Highest accuracy after fine-tuning

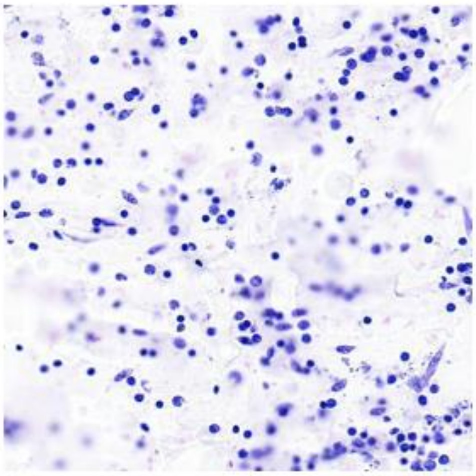
Table 4.5 Execution Time Comparison on Complete Dataset

Model	Execution Time	Observation
BDA	34 hours	Accurate contour output but higher processing time
MEDIAR	16 hours	Faster than BDA and suitable for segmentation
YOLOv8/F-T	68 minutes	Fastest among the compared models

4.3.2 Comparison with Existing Methods

Existing approaches in biomedical image analysis often use either a single segmentation model or traditional image processing techniques. Such systems may not provide enough flexibility when images contain dense cellular regions, overlapping cells, or varying image quality. Manual microscopy analysis is also slow and depends heavily on expert observation.

The proposed framework improves upon these limitations by combining multiple deep learning models in a unified system. YOLOv8 is used for fast detection, BDA is used for contour-based cell boundary extraction, and MEDIAR is used for microscopy segmentation. This multi-model structure allows better comparison and makes the system more useful for research-based brain cell analysis.



(a)

(b)

Figure 4.4 BDA Contour Detection Output

Table 4.6 Comparison of 5 Existing Models

Existing Method	**Limitation**	**Proposed Improvement**
Manual microscopy analysis	Time-consuming and error-prone	Automated cell detection workflow
Traditional segmentation	Requires manual preprocessing	Deep learning-based segmentation
Single-model framework	Limited flexibility	Multi-model integration
Conventional detection	Weak for complex cellular structures	YOLOv8-based detection
Basic contour extraction	Sensitive to noise	BDA-based contour proposal approach
Large image direct processing	High memory usage	MEDIAR tiled segmentation approach

4.4 Significance of the Results Obtained

The obtained results demonstrate the effectiveness of deep learning-based segmentation and object detection techniques for automated brain cell analysis. The framework successfully reduces the need for manual detection and improves the efficiency of microscopy image analysis. The integration of multiple state-of-the-art models allows researchers to compare outputs and select suitable approaches for different image characteristics.

The proposed framework also supports high-resolution image processing through slicing, tiling, and contour-based extraction methods. The generated outputs provide meaningful information regarding cellular structures and boundaries, which can support biomedical research and neurological analysis. The successful implementation of the framework demonstrates the practical applicability of deep learning in medical image processing tasks.

4.5 Deviations from Expected Results and Justification

Although the framework generates meaningful segmentation and detection outputs, certain deviations from expected results were observed during implementation. Some models produced inaccurate detections in densely packed cellular regions due to overlapping structures and irregular boundaries. Variations in image quality and illumination also affected segmentation consistency in certain cases.

High computational requirements and GPU memory limitations affected processing speed for large microscopy images. Some segmentation models required additional preprocessing and patch reconstruction steps for better output quality. Detection models generated faster outputs but sometimes could not capture fine cellular boundaries compared to segmentation approaches.

Such deviations are expected since the analysis of microscopy images is complex and the model architectures are different. Further optimization, expansion of the dataset, and fine-tuning can improve the overall performance of the framework.

4.6 Social and Environmental Impact

The proposed automated brain cell detection framework has a positive societal impact in the area of biomedical research and healthcare technology. This removes the need for human effort in microscopy image analysis, allowing researchers to analyze complex cellular structures more efficiently. The automation of the analysis can enhance research productivity and decrease the time required to process images at a large scale.

The suggested system is designed in a digital way from the environmental perspective and does not include any dangerous physical processes or hazardous substances. With cloud-based execution environments like Google Colab, large dedicated hardware infrastructure is less necessary. Besides, efficient deep learning-based analysis could reduce redundant manual processing and enhance the use of computational resources. The project promotes intelligent biomedical imaging systems and supports the future development of AI-assisted healthcare technologies.

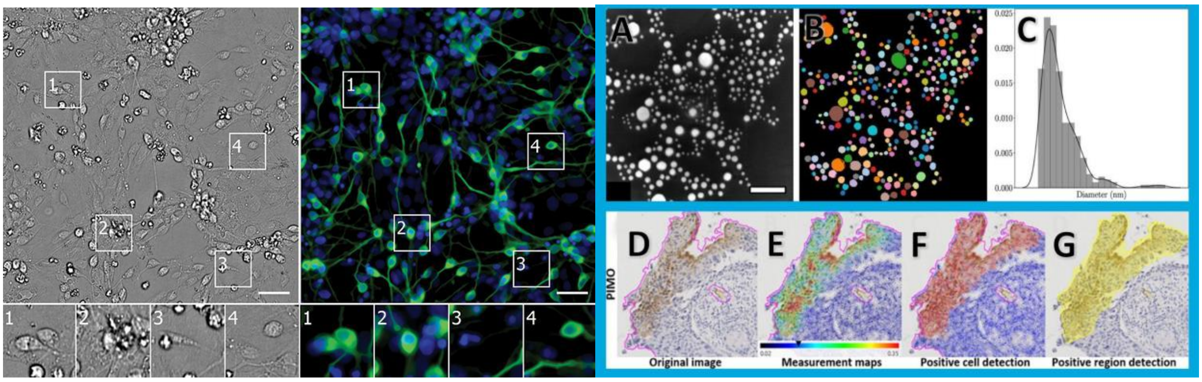


Figure 4.2 Output Visualization of Brain Cell Detection

4.7 Summary and Conclusions

The experimental results and analysis of the proposed automated brain cell detection framework were explained in this chapter. The outputs of the segmentation and detection models showed the potential of deep learning techniques to analyse complex brain microscopy images.

The comparative evaluation pointed out the respective strengths and limitations of the different models embedded in the framework. The results obtained show that the combination of several segmentation and detection approaches leads to increased flexibility and efficient biomedical image analysis. The environmental and societal analysis also highlights the significance of intelligent AI-based systems in future healthcare and research applications.

CHAPTER 5
CONCLUSION AND FUTURE SCOPE

5.1 Brief Summary of the Work

The proposed project focuses on automated brain cell detection and segmentation using advanced deep learning techniques. Brain microscopy image analysis is a challenging task due to overlapping cellular structures, varying image intensity, and complex tissue patterns. Manual analysis of such images is time-consuming and prone to human error, creating the need for automated and intelligent biomedical image processing systems. The primary objective of this project is to develop a unified framework capable of integrating multiple segmentation and object detection models for efficient brain cell analysis.

The methodology adopted in this project involves image preprocessing, model execution, post-processing, and result visualization within a unified framework. High-resolution brain microscopy images are processed using deep learning architectures such as SAM, Mask R-CNN, CA-Net, SAHI, DETR, BDA, MEDIAR, and YOLOv8. Segmentation models generate pixel-level masks representing cellular boundaries, while detection models identify cellular regions using bounding boxes. The system also includes a graphical user interface and backend integration to simplify model execution and result visualization.

The implemented framework supports comparative analysis between different segmentation and detection approaches. Preliminary implementation results demonstrate that deep learning models can effectively identify and analyse complex cellular structures from microscopy images. The proposed framework improves automation, reduces manual effort, and provides a scalable research-oriented solution for biomedical image analysis.

5.2 Conclusions

The project successfully demonstrates the application of deep learning techniques for automated brain cell detection and segmentation. The integration of multiple state-of-the-art models within a unified framework improves flexibility and allows systematic comparison of segmentation and detection approaches. Models such as SAM and Mask R-CNN provide detailed segmentation outputs, while YOLOv8 offers faster object detection and efficient localization of cellular structures.

The obtained results indicate that deep learning-based biomedical image analysis can significantly improve the efficiency of microscopy image processing tasks. The framework successfully processes high-resolution brain microscopy images and generates meaningful outputs such as segmentation masks and detection bounding boxes. The use of preprocessing techniques such as slicing, tiling, normalization, and contour extraction further improves model performance and image analysis quality.

The proposed system also demonstrates the importance of combining frontend visualization with backend deep learning execution. The GUI-based framework simplifies experimentation and allows researchers to visualize outputs effectively. Overall, the project establishes a strong foundation for automated biomedical image analysis using advanced segmentation and detection architectures.

5.3 Significance of the Results Obtained

The results obtained from the proposed framework demonstrate the practical applicability of deep learning in biomedical image analysis. Automated brain cell detection reduces manual effort and improves analysis efficiency for researchers working with microscopy images. The integration of multiple segmentation and detection models enables comparative evaluation and helps identify suitable approaches for different image characteristics.

The framework also supports high-resolution image processing through advanced techniques such as tiled segmentation and contour-based extraction. The generated outputs provide meaningful information regarding cellular structures and tissue organization, which can support future neurological and biomedical research applications. The successful implementation of the framework highlights the potential of artificial intelligence in improving healthcare-related image analysis systems.

The proposed methodology is a step towards scalable and intelligent solutions for medical imaging. The system provides a research platform for evaluating modern deep learning architectures in the context of biomedical imaging.

5.4 Future Work Scope

The proposed framework can be extended in the future by incorporating more deep learning architectures for better segmentation and object detection performance. Advanced transformer-based models and foundation models can be integrated to improve the boundary extraction and cellular feature representation. Fine-tuning on larger annotated datasets can further improve the accuracy and robustness of the model for complex microscopy images.

Another important future extension is to develop a fully integrated real-time web application for biomedical image analysis. The existing GUI and backend architecture can be scaled up to a cloud-based deployment platform for large-scale dataset handling and remote experimentation. Data accessibility for researchers and healthcare professionals can be improved with real-time processing and visualization features.

Future work may also include improving the efficiency of the proposed method and reducing the GPU memory cost in processing high-resolution images. Optimized preprocessing techniques, lightweight model architectures, and distributed execution techniques can improve scalability and the speed of execution. The framework can be extended to other medical imaging applications like tissue analysis, disease detection, and cellular classification tasks.

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CO and PO Mapping

NBA:

Table A1.1 Course Articulation Matrix

CO		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CSE 4299.1	Apply mathematics, science and engineering skills to identify, formulate, synthesize and solve the problems from various areas of computer science engineering.	3	3	2	2	2	-	-	-	-	-	-	2	3	2	2	-
CSE 4299.2	Have knowledge of new trends in engineering/technology by developing programmable coding in various computer programming languages.	2	2	3	2	3	-	-	-	-	-	-	3	3	3	2	2
CSE 4299.3	Use the industry standard tools to analyze, design, develop and test software engineering based applications.	2	2	3	3	3	-	-	-	1	2	1	2	3	3	2	2
CSE 4299.4	Apply theoretical knowledge to real-world engineering problems and manage complex engineering projects.	3	3	3	3	2	1	-	-	2	2	2	2	3	2	3	2
CSE 4299.5	Acquire skills of collaboration and independent learning.	-	-	1	1	1	-	-	-	3	3	2	3	1	2	2	2
CSE 4299 (Avg. correlation level)		**2.0**	**2.0**	**2.4**	**2.2**	**2.2**	**0.2**	**-**	**-**	**1.2**	**1.4**	**1.0**	**2.4**	**2.6**	**2.4**	**2.2**	**1.6**

PROGRAM OUTCOMES (PO)

Engineering graduates will be able to:

PO1. Engineering Knowledge:
Apply the knowledge of mathematics, science, engineering fundamentals, and artificial intelligence concepts to solve complex problems in brain cell detection and biomedical image analysis.

PO2. Problem Analysis:
Identify, formulate, review research literature, and analyze complex problems related to brain microscopy image segmentation and object detection using deep learning techniques.

PO3. Design and Development of Solutions:
Design and develop intelligent solutions for automated brain cell detection using segmentation and object detection models with consideration for healthcare and biomedical research applications.

PO4. Conduct Investigations of Complex Problems:
Use research-based methodologies, experimental analysis, preprocessing techniques, and comparative evaluation to study the performance of different deep learning models.

PO5. Modern Tool Usage:
Apply modern engineering and AI tools such as Python, OpenCV, PyTorch, TensorFlow, YOLOv8, React, Flask, and Google Colab for developing and testing biomedical image analysis systems.

PO6. The Engineer and Society:
Apply engineering solutions responsibly to support biomedical research and improve automated analysis systems for healthcare-related applications.

PO7. Environment and Sustainability:
Understand the societal and environmental benefits of digital biomedical image analysis systems that reduce manual effort and support efficient computational processing.

PO8. Ethics:
Apply ethical principles and maintain professional responsibility while developing AI-based biomedical imaging solutions and handling research data.

PO9. Individual and Team Work:
Function effectively as an individual and collaborate in multidisciplinary environments during research, implementation, and documentation of the project.

PO10. Communication:
Communicate technical concepts, experimental results, and project outcomes effectively through reports, presentations, graphical outputs, and documentation.

PO11. Project Management and Finance:
Apply project planning, scheduling, and management principles for systematic execution of the proposed brain cell detection framework.

PO12. Life-long Learning:
Recognize the importance of continuous learning and adapting to emerging technologies in artificial intelligence, computer vision, and biomedical image analysis.

PROGRAM SPECIFIC OUTCOMES (PSO)

PSO1.
Analyze and solve real-world biomedical image analysis problems by applying deep learning models, image processing techniques, and software-based implementation frameworks for automated brain cell detection.

PSO2.
Formulate and develop optimized solutions for computationally intensive microscopy image processing applications using advanced segmentation and object detection architectures such as YOLOv8, BDA, MEDIAR, and SAM.

PSO3.
Design and model intelligent biomedical image analysis applications using standard software engineering practices, graphical user interfaces, backend integration, and modular deep learning workflows.

PSO4.
Design and develop scalable solutions for distributed and cloud-based processing using frameworks such as Flask, Google Colab, and GPU-enabled deep learning environments for efficient brain image analysis.

Table A1.2 Program Articulation Matrix

COURSE CODE	Course Title	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PS02	PSO3	PSO4
CSE 4299	Project Work	**2.0**	**2.0**	**2.4**	**2.2**	**2.2**	**0.2**	-	-	**1.2**	**1.4**	**1**	**2.4**	**2.6**	**2.4**	**2.2**	**1.6**

IET (AHEP Mapping):

Table A1.3 CLO-AHEPLO Mapping

		AHEP LOs						
CLOs	**Statements**	**C8**	**C9**	**C10**	**C12**	**C15**	**C16**	**C17**
1	Apply mathematics, science and engineering skills to identify, formulate, synthesize and solve the problems from various areas of computer science engineering.	1	2	1	3	1	1	1
2	Have knowledge of new trends in engineering/technology by developing programmable coding in various computer programming languages.	1	1	2	3	2	1	2
3	Use the industry standard tools to analyze, design, develop and test software engineering based applications.	1	2	2	3	2	2	2
4	Apply theoretical knowledge to real-world engineering problems and manage complex engineering projects.	2	3	2	2	3	2	2
5	Acquire skills of collaboration and independent learning.	1	1	1	1	2	3	3

AHEPS

C8	Identify and analyse ethical concerns and make reasoned ethical choices informed by professional codes of conduct.
C9	Use a risk management process to identify, evaluate and mitigate risks associated with a particular project or activity.
C10	Adopt a holistic and proportionate approach to the mitigation of security risks.
C12	Use practical laboratory and workshop skills to investigate complex problems.
C15	Apply knowledge of engineering management principles, commercial context, project and change management, and relevant legal matters including intellectual property rights.
C16	Function effectively as an individual, and as a member or leader of a team.
C17	Communicate effectively on complex engineering matters with technical and non-technical audiences.

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22	P. Hema Rao, Badri Vishal Padamwar. "Mathematical Methods in Image Processing: A Survey", Turkish Journal of Computer and Mathematics Education (TURCOMAT), 2019 Publication	<1%
23	umpir.ump.edu.my Internet Source	<1%
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26	H.L. Gururaj, Francesco Flammini, J. Shreyas. "Data Science & Exploration in Artificial Intelligence", CRC Press, 2025	<1%

PROJECT DETAILS

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Project Details			
Project Title	**Automated Brain Cell Detection using Deep Learning Techniques**		
Project Duration	4 Months	Date of reporting	20-5-2026
Organization Details			
Organization Name	**Manipal Institute of Technology (MIT), Bengaluru**		
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